SAHAYAK Presentation Guide

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Role: Introduction, Problem and Project Purpose

Suggested speaking time: 3 to 4 minutes

Your job

You are the first speaker. Explain why the project matters, introduce SAHAYAK, describe its goal and limits, then hand over to Shyamantika. You do not need to explain every technical detail.

Slide 1 - Opening

Say:

Good morning respected judges, teachers, and everyone here. We are presenting SAHAYAK, a real-time Indian Sign Language recognition prototype. Our project explores how a camera and a computer program may help recognize a limited set of trained signs. We will explain the problem, show how our prototype works, and talk honestly about what it can and cannot do.

Show: the homepage and the project name. Pause, look at the judges, and introduce the team.

Slide 2 - The communication challenge

Say:

People who use sign language may find it hard to communicate with someone who does not understand their signs. This can happen in classrooms, public places and daily conversations. Technology may help in some situations, but it must be designed carefully and tested with the people who may use it.

Indian Sign Language is a full language. It has its own rules and ways of showing meaning. Recognizing a few hand signs is not the same as understanding a full conversation.

Slide 3 - Why accessibility matters

Optional fact: The 2011 Census of India counted 5,071,007 people in its hearing-disability category. This is an old census figure. It is not the number of ISL users and it is not a current estimate.

Say:

This figure shows that hearing accessibility is an important topic, but we must not confuse hearing-disability figures with the number of people who use ISL. Our project explores one small part of a much bigger issue.

Source: https://censusindia.gov.in/nada/index.php/catalog/43369/

For current global hearing-loss information, check: https://www.who.int/news-room/fact-sheets/detail/deafness-and-hearing-loss

Slide 4 - What SAHAYAK does

Say:

SAHAYAK uses a camera to see a hand pose or movement. The website finds key points on the hand and changes those points into numbers. A trained model uses the numbers to guess one of the signs it has learned. The website then displays the predicted label as text.

A landmark is a key point on the hand, such as a fingertip or joint. A model learns patterns from examples that have labels. It does not understand a sign exactly as a person does.

Slide 5 - Goal and limits

Say:

Our goal is to show a working path from camera input to a predicted sign label. The current system supports a limited set of trained signs. It is not a complete ISL translator and cannot replace a qualified interpreter.

The website also has extra speech and translation features, but not every feature works offline. Results can change with lighting, camera angle, hidden hands, movement and differences between

signers.

Handover to Shyamantika

Say:

I have explained the problem, why accessibility matters and what we hope to show. My teammate Shyamantika Pegu will now explain how the system works, how hand points become numbers and how the trained model makes a prediction.

Look toward Shyamantika and pause before handing over.

Questions judges may ask

Q: Is sign language the same around the world?

A: No. Different sign languages exist, and each has its own rules.

Q: Does SAHAYAK translate every ISL sentence?

A: No. It recognizes a limited set of trained labels. Full sign-language translation is much harder.

Q: Can it replace an interpreter?

A: No. It is an educational prototype, not a substitute for a trained interpreter.

Q: Why is it useful if it has limits?

A: It lets us explore an assistive-technology idea, learn about recognition and test how it can improve.

Q: Does it always work offline?

A: No. Live predictions and some translation or speech features need internet access.

Practice checklist

- Practise the opening and handover aloud.
- Speak clearly and look at the judges instead of reading every word.
- Use the census number only with its correct meaning.
- Do not claim an accuracy percentage unless a real test report supports it.
- If asked about something you do not know, say you will verify it rather than guessing.

One-sentence summary

I explain why SAHAYAK was built, what problem it explores, and why it is a limited sign-recognition prototype rather than a full sign-language translator.